

Module- 1

Notes on Quantum Chemistry

Schrodinger wave equation – Derivation from classical wave equation.

Let us consider the simplest type of wave motion – the vibration of a stretched string. If ω is the amplitude of vibration at any point x at a time t , then the partial differential equation for wave motion is

$$\frac{\partial^2 \omega}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 \omega}{\partial t^2} \quad \text{-----(1)}$$

Where v is the velocity of the wave, the value of ω depends on two variables x and t . To separate the variables let us consider $\omega = f(x) g(t)$ -----(2)

where $f(x)$ and $g(t)$ are functions of coordinate x and time t respectively. For stationary (standing waves) such as occurring in stretched string $g(t)$ may be represented as

$$g(t) = A \sin(2\pi vt) \quad \text{----- (3)}$$

Where v is the vibrational frequency and A is a constant which is known as the maximum amplitude. Thus equation (2) becomes

$$\omega = f(x) A \sin(2\pi vt) \quad \text{----- (4)}$$

Differentiating w.r.t time we get

$$\frac{\partial^2 \omega}{\partial t^2} = -4\pi^2 v^2 f(x) g(t) \quad \text{----- (5)}$$

Combination of (1) and (5)

$$\frac{\partial^2 \omega}{\partial x^2} = \frac{-4\pi^2 v^2}{v^2} f(x) \quad \text{----- (6)}$$

The frequency of vibration and velocity v are related by the expression $V = v\lambda$ where λ is the wave length. Hence (6) becomes

$$\frac{\partial^2 \omega}{\partial x^2} = \frac{-4\pi^2}{\lambda^2} f(x) \quad \text{----- (7)}$$

This equation is the time independent wave equation and suitable for setting up a wave function to describe wave mechanics

The Schrodinger wave equation.

The Schrodinger wave equation is formulated by combining pure mathematical property of waves and the wave property of matter.

The equation (7) represents wave motion in one direction only. Replacing to Cartesian co-ordinates it becomes

$$\frac{\partial^2 \Psi}{\partial x^2} + \frac{\partial^2 \Psi}{\partial y^2} + \frac{\partial^2 \Psi}{\partial z^2} = \frac{-4\pi^2}{\lambda^2} \Psi \quad \text{----- (7)}$$

②

Where Ψ is a function of Cartesian coordinates x, y, z . Replacing for Laplacian operator ∇^2

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

We have $\nabla^2 \Psi = \frac{-4\pi^2}{\lambda^2} \Psi$ ----- (8)

Upon introducing de Broglie relationship $\lambda = h/mv$, we get

$$\nabla^2 \Psi = \frac{-4\pi^2 m^2 v^2}{h^2} \Psi$$
 ----- (9)

Where m is the mass and v is the velocity of the particle. The total energy E of the particle is the sum of kinetic energy $\frac{1}{2} mv^2$ and potential energy V

$$E - V = \frac{1}{2} mv^2, \quad \therefore v^2 = \frac{2(E-V)}{m}$$

Substituting in (9), $\nabla^2 \Psi = \frac{-8\pi^2 m (E-V)}{h^2} \Psi$ ----- (10)

Or $\frac{\partial^2 \Psi}{\partial x^2} + \frac{\partial^2 \Psi}{\partial y^2} + \frac{\partial^2 \Psi}{\partial z^2} + \frac{8\pi^2 m}{h^2} (E - V) \Psi = 0$ ----- (11)

Multiplying by $-\frac{h^2}{8\pi^2 m}$ and rearranging, we get

$$-\frac{h^2}{8\pi^2 m} \left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right] \Psi + V \Psi = E \Psi$$
 ----- (12)

$$\left[\frac{-h^2}{8\pi^2 m} \nabla^2 + V \right] \Psi = E \Psi$$
 ----- (13)

Where $\left[\frac{-h^2}{8\pi^2 m} \nabla^2 + V \right]$ is called Hamiltonian operator $\hat{H}$, then equation (12) becomes

$$\hat{H} \Psi = E \Psi$$
 ----- (14)

Equations (10), (11), (12), (13), and (14) are the different forms of **Schrodinger wave equation**. This equation has been used to solve a number of problems related to atomic and molecular structures.

Schrodinger wave equation describes the behaviour of electron wave in atoms and molecules. The equation is time independent wave equation and so applies only to stationary states. The Hamiltonian operator $\hat{H}$ in Schrodinger equation consists of two parts, kinetic energy part $\frac{-h^2}{8\pi^2 m} \nabla^2$ and the potential energy part (V).

Significance of Ψ (wave function) : The wave function Ψ in Sch.eqn show how the wave amplitude varies with distances along x, y, z - axes and to derive solutions of differential equation.

Therefore only those mathematical function which satisfy certain conditions are acceptable and are called **eigenfunctions** of the system and the energies E corresponding to these eigen functions are called **eigen values**. If an operator $\hat{A}$ operating on a function gives back function itself multiplied by a constant, then the function is referred to as an eigen function of the operator and the constant is the associated eigen value.

The conditions which a wave function Ψ must satisfy before it is acceptable are:

1. **Wave function Ψ has to be single valued** :- This says that at any given point in x,y,z-direction(coordinates) it can have only one value. For ex, let us consider $\sin^{-1} x$, if we put $x = 0$ $\sin^{-1} x$ can take values $0, \pi, 2\pi, 3\pi \dots n\pi$. ie the wave function has infinite number of values for a given value of x which means probability density will also have more than one value. Hence this function is not acceptable.
2. **Wave function Ψ has to be continuous**:-If the function is discontinuous probability will also change discontinuously
3. **Wave function Ψ has to be normalized**:- As the system is definitely in space, the **probability** of finding the system anywhere in space is unity. Therefore Ψ has to satisfy the condition

$$\iiint_{-\infty}^{+\infty} \Psi^* \Psi \, dx \, dy \, dz = 1$$

or simply as $\int_{-\infty}^{+\infty} \Psi^* \Psi \, d\tau = 1$ is the normalization condition or probability condition if Ψ is a complex function. Ψ^* is the complex conjugate of Ψ . or

$$\int_{-\infty}^{+\infty} \Psi^2 \, d\tau = 1 \text{ if } \Psi \text{ is a real function}$$

If Ψ is not normalized it is possible to normalize it by multiplying the function with a suitable constant.

Application of Schrodinger equation to 1-D (one dimensional) box solutions.

Let us consider a particle of mass m is allowed to move only along x-axis in forward and backward directions. The length of the box is 'a' and height is infinity. So potential energy (V) is infinity ($V = \infty$) on the walls and outside the box. Within the box. $V = 0$ Therefore, the wave function Ψ and probability of finding the particle (Ψ^2) must be zero when $x \leq 0$ and $x \geq a$ i.e., particle does not exist outside the box. Schrodinger equation for this is

Region-1	Region-11	Region-111
$x \leq 0$	$0 < x < a$	
$V = \infty$	$V = 0$	
$x = 0$	$x = a$	

$$\frac{\partial^2 \Psi}{\partial x^2} + \frac{8\pi^2 m}{h^2} E \Psi = 0 \quad \text{-----(1)}$$

(particle is moving in x-direction only and within the box $V=0$)

$$\text{Let us put } \frac{8\pi^2 m E}{h^2} = k^2, \text{ and } \hbar = \frac{h}{2\pi}, \therefore \hbar^2 = \frac{h^2}{4\pi^2}$$

$$\text{Or } \frac{2mE}{\hbar^2} = k^2$$

Then eqn (1) becomes

$$\frac{\partial^2 \Psi}{\partial x^2} + k^2 \Psi = 0 \quad \text{-----(2)}$$

General solution for this equation is

$$\Psi = A \sin(kx) + B \cos(kx) \quad \text{-----(3)}$$

Since by the boundary condition, at $x = 0$, $\Psi = 0$

We get $\Psi_{(0)} = A \sin(0) + B \cos(0)$ which implies that $B = 0$

$$\therefore \text{The solution becomes } \Psi_{(x)} = A \sin(kx) \quad \text{-----(4)}$$

When we apply the next boundary condition, at $x = a$ $\Psi_{(a)} = 0$

$$\text{Then } \Psi_{(a)} = A \sin(ka) = 0 \quad \text{-----(5)}$$

This means $ka = n\pi$ where $n = 0, 1, 2, 3, \dots$

$$\text{Or } k = \frac{n\pi}{a}$$

Thus wave function (5) becomes $\Psi_{(a)} = A \sin\left(\frac{n\pi x}{a}\right)$

But when $n = 0$, $\Psi_{(x)} = 0$ $\therefore$ we restrict n to values $1, 2, 3, \dots$

We have

$$k^2 = \frac{2mE}{\hbar^2} = \frac{n^2 \pi^2}{a^2}$$

$$\therefore E = \frac{\hbar^2 n^2 \pi^2}{2ma^2} \quad \text{or} \quad E = \frac{n^2 h^2}{8ma^2} \quad \text{-----(6)}$$

where n is an integer having values $1, 2, 3, \dots$ etc. It can never be zero or fractional. i.e. a free particle when confined to move in a certain range of space, the energy values become quantized. These quantized energy values are called **eigen values**.

when $n = 1$, $E_1 = \frac{h^2}{8ma^2}$ is the lowest energy of the particle in 1-D box and is called **zero point energy**.

$$n = 2, E_2 = \frac{4h^2}{8ma^2}, \quad n = 3, E_3 = \frac{9h^2}{8ma^2} \text{ etc.}$$

and energy function of n is $\Psi_{(a)} = A \sin\left(\frac{n\pi x}{a}\right)$

The value of A is determined from **normalization condition**

$$\int_{-\infty}^{+\infty} \Psi^2 d\tau = 1 = \int_0^a A^2 \sin^2\left(\frac{n\pi}{a} x\right) dx = 1$$

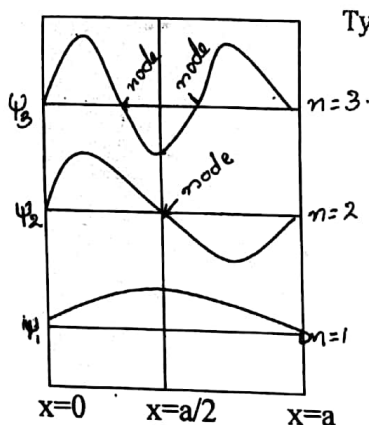
Which on solving gives $A = \sqrt{\frac{2}{a}}$, The value of constant A is known as **normalization factor**.

Therefore, the acceptable wave functions or normalized wave functions of Schrodinger's wave equation for a particle in a one-dimensional box could be written as

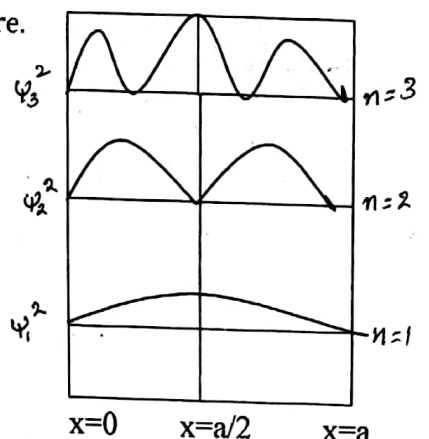
$$\Psi_{(n)} = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a} x\right)$$

$$\text{Or } \Psi_{(n)} = \sqrt{\frac{2}{a}} \sin\left(\sqrt{\frac{8\pi^2 m E}{h^2}} x\right) \quad \text{-----(7)}$$

The plots of $\Psi_{(n)}$ and Ψ_n^2 against x are shown in fig.



plot of Ψ against x



plot of Ψ^2 against x

The features of above figures:

The figures show that (1) the quantum number, n represents the nodes in particle wave. A node is a point where the probability of finding the particle is zero. In the first energy level there is no internal node, in the second energy level there is one node and in third level there are two nodes. Thus in general there are $(n-1)$ nodes in the particle wave.

(2). For a potential box of fixed size, as the number of nodes increases, the wave length decreases, hence the energy of the level increases, the total potential energy has been assumed to be zero.

(3). It can be seen that the value of Ψ may be either positive or negative, but Ψ^2 being the probability of finding the particle at any point is always positive.

④: The probability density is not uniform at all positions. When n is small, the probability density is more pronounced. When n is increased, the distribution becomes more and more uniform. For very high values of n , the distribution is completely uniform. Thus on an average, the particle spends equal amounts of time at all points. Thus classical behaviour emerges when n becomes very large. This fact is in agreement with correspondence principle which says that the predictions of quantum mechanics for a very large value of n are identical to those predicted by classical mechanics.

An electron is confined to move in a 1-D box of length A^0 . Calculate its ground state energy in electron volts (eV). Calculate the energy of first excited state.

$$\text{Ground state energy } (n=1) \quad E_1 = \frac{n^2 h^2}{8ma^2} = \frac{1^2 (6.625 \times 10^{-35} \text{ Js})^2}{8(9.11 \times 10^{-31} \text{ kg}) (10^{-10} \text{ m})^2 (1.602 \times 10^{-19} \text{ J/eV})} = 37.6 \text{ eV}$$

For the first excited state $n = 2$

$$E_2 = \frac{2^2 (6.625 \times 10^{-35} \text{ Js})^2}{8(9.11 \times 10^{-31} \text{ kg}) (10^{-10} \text{ m})^2 (1.602 \times 10^{-19} \text{ J/eV})} = 150.4 \text{ eV}$$

Application of 1-D box solution to conjugated molecules.

(refer : Engg chemistry by Shashi chowla)

6

Forms of hydrogen atom wave functions.

In the case of atoms the potential energy field is spherically symmetrical (V depends only on r). So it is convenient to transform the Schrodinger equation (x, y, z) into the spherical polar coordinates r, θ and ϕ by using the relations

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

we can write the wave function Ψ as

$$\Psi_{r, \theta, \phi} = R(r) \Theta(\theta) \Phi(\phi)$$

Where R is a function depends only on r , Θ is the function that depends on θ , Φ function depends on ϕ .

The transformed Schrodinger equation in spherical polar coordinates for the hydrogen-like species can be separated into three wave function equations as given below.

(1) Function involving only r .

$$\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{8\pi^2 \mu r^2}{h^2} \left(E + \frac{Ze^2}{(4\pi\epsilon_0)r} \right) = l(l+1) \quad \text{---- (1)}$$

(2) Function involving only θ

$$\frac{\sin \theta}{\Theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + l(l+1) \sin^2 \theta = m^2 \quad \text{---- (2)}$$

(3) Function involving only ϕ

$$\frac{1}{\Phi} \frac{d^2 \Phi}{d\phi^2} = -m^2 \quad \text{---- (3)}$$

Solutions of the three functions :

1. The allowed values of energy of the electron can be obtained from equation (1) as

$$E = - \frac{2\pi^2 \mu Z^2 e^4}{n^2 h^2 (4\pi\epsilon_0)^2} \quad \text{where } n \text{ is referred to as the principal quantum number } n=1,2,3,\dots\text{etc}$$

We conclude that the energy depends only on $R(r)$ the distance of electron from the nucleus and not on its angular parts. The energy levels are quantized and depends on n .

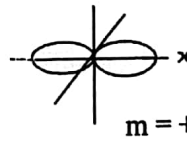
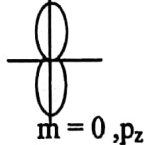
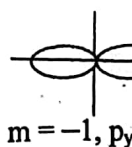
2. Equation(2) yields the angular momentum of the motion of electron represented by the angular momentum quantum number or azimuthal quantum number l having values $0, 1, 2, \dots, (n-1)$ The angular momentum associated with a particular value of l is given by $\sqrt{l(l+1)} \left(\frac{h}{2\pi} \right)$. This function is generally named as *atomic orbital*. The plot of the function gives the shape of atomic orbitals. The wave function corresponding to $l = 0$ is called s orbital, $l = 1$ are called p orbitals and $l = 2$ as d orbitals.

3. The solution of equation (3) gives m called magnetic quantum number, $m = 0, \pm 1, \pm 2, \pm 3 \dots \pm l$. This gives the direction of atomic orbitals in space. Polar representation of s and p electrons of hydrogen is given below:

$l = 0, m = 0$ s -orbital

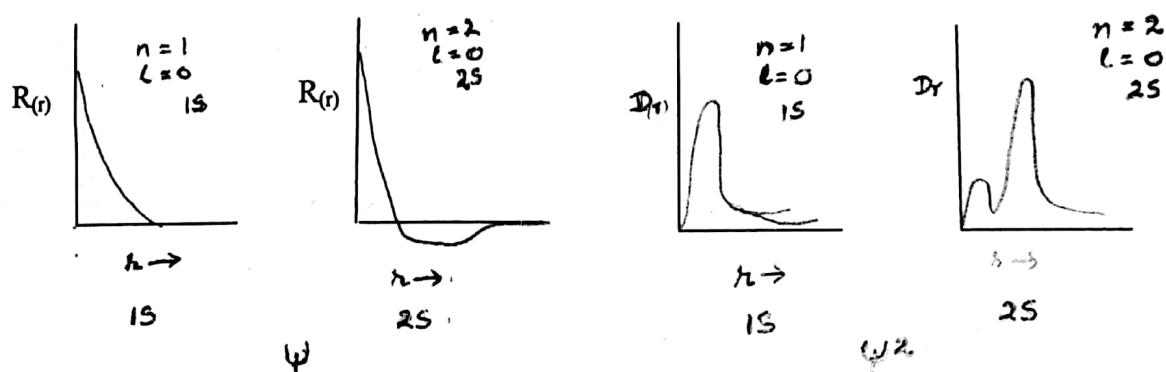


$l = 1$

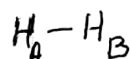


$m = +1, p_x\text{-orbital}$

Plot of radial part $R(r)$ of hydrogen atom wave function for 1s and 2s orbitals.



Energy level diagrams of diatomic molecules.



Schrodinger wave equation for a system containing more than one electron is difficult. In such cases approximate methods are used. In molecular orbital theory (MOT) the wave function is generally taken as a linear combination of atomic orbitals (LCAO). Let us consider a diatomic molecule. The wave function for an electron moving in the field of two nuclei A and B is

$$\Psi = C_A \Psi_A + C_B \Psi_B$$

Where Ψ_A and Ψ_B are atomic orbital wave functions of A and B, C_A and C_B are coefficients chosen to give lowest energy for the molecular orbitals (MO's).

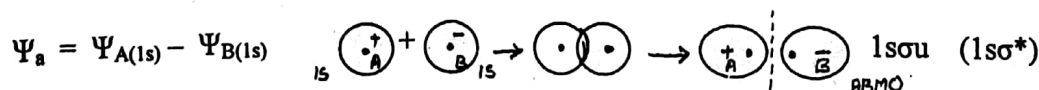
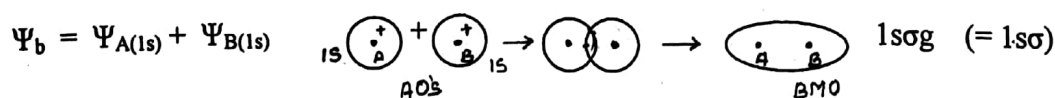
In the case of homonuclear diatomic molecules ($H_2, O_2, \dots$) equal weights are given to Ψ_A and Ψ_B on symmetry ground, so that $C_A^2 = C_B^2$ or $C_A = \pm C_B$. If $C_A = 1$, $C_B = \pm 1$ Hence we have two MO's

$$\Psi_+ = \Psi_A + \Psi_B \quad \text{and} \quad \Psi_- = \Psi_A - \Psi_B$$

Thus when two atomic orbitals combine together, two molecular orbitals are formed one is of lower energy called bonding MO and the other of higher energy called antibonding MO. (+) sign indicates the electron density in the region between the nuclei, and (-) sign leads to decrease of electron density in this region. For hydrogen molecule the atomic orbitals are 1s from each H atoms. The bonding and antibonding MO's are then

$$\Psi_b = H_{A(1s)} + H_{B(1s)} \quad \text{and} \quad \Psi_a = H_{A(1s)} - H_{B(1s)}$$

The MO's can be represented as



The positive sign leads to a buildup of electron density in the region between the nuclei, whereas the negative sign leads to the depletion of electron density in this region. The subscripts g and u

refer to the symmetry of molecular orbitals. g represents that MO is symmetric with respects to the inversion centre and u means antisymmetric w.r.t the inversion centre.

In **poly atomic systems**, when two atomic orbitals combine together to form two molecular orbitals(MO's) by a symmetric combination are designated as σ and by the unsymmetric combination are named as π orbitals. Thus MO's in the order of their energies can be arranged as $1s\sigma < 1s\sigma^* < 2s\sigma < 2s\sigma^* < 2p_x\sigma < 2p_x\pi = 2p_y\pi < 2p_x\pi^* = 2p_y\pi^* < 2p_z\sigma^*$

All atomic orbitals of interacting atoms are combined in pairs and must of same energy and must have same symmetry. The MO's formed are arranged in ascending order of energy. All the electrons in AO's (atomic orbitals) are redistributed in MO's. based on aufbau principle and Pauli's exclusion principle. Thus maximum number of electrons in each MO is two. The bonding orbitals are filled first than antibonding orbitals.

For example energy level diagram of O_2 molecule can be given as

$$1s\sigma^2 < 1s\sigma^{*2} < 2s\sigma^2 < 2s\sigma^{*2} < 2p_x\sigma^2 < 2p_x\pi^2 = 2p_y\pi^2 < 2p_x\pi^1 = 2p_y\pi^1 < 2p_z\sigma^{*2}$$

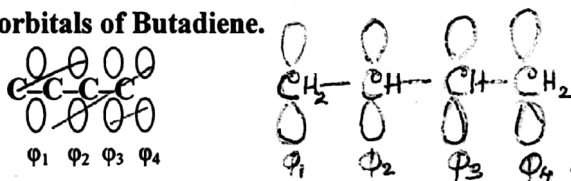
This arrangement explains that O_2 is paramagnetic. $1s\sigma^2 < 1s\sigma^{*2} < 2s\sigma^2 < 2s\sigma^{*2} < 2p_x\pi_u^2 = 2p_y\pi_u^2 < 2p_x\pi_g^1 = 2p_y\pi_g^1 < 2p_z\sigma_g^{*2}$

Q : Write down the energy level diagram of N_2 and F_2 .

$$2p_x\pi_g^2 = 2p_y\pi_g^2 < 2p_x\pi_u^1 = 2p_y\pi_u^1 < 2p_z\sigma_g^{*2}$$

MOT of conjugated systems

Pi molecular orbitals of Butadiene.



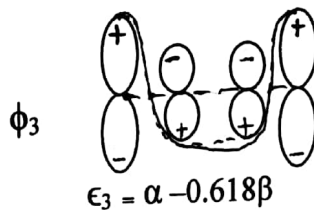
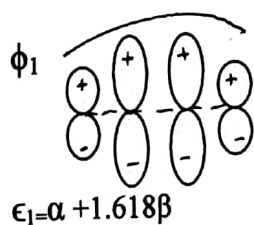
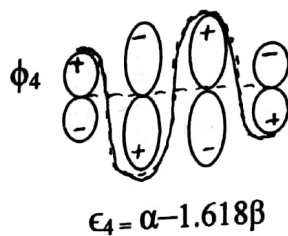
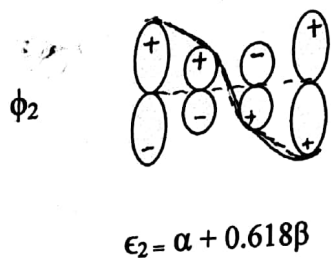
The trial function is $\phi = C_1\phi_1 + C_2\phi_2 + C_3\phi_3 + C_4\phi_4$

Carbon atom is in sp^2 hybridisation. Apart from σ frame work, π electrons move independently. Schrodinger equation can be applied to calculate approximate energy of pi molecular orbitals. In variation method, the wave function is generally taken as a linear combination atomic orbital wave functions multiplied by the coefficients $C_1, C_2, C_3, \dots$. The coefficients $C_1, C_2, C_3, \dots$ are chosen such that when used in equation (1) gives minimum value of energy.

$$\frac{\int_{-\infty}^{+\infty} \phi H \phi d\tau}{\int_{-\infty}^{+\infty} \phi \phi d\tau} \geq E_0 \quad \dots\dots (1)$$

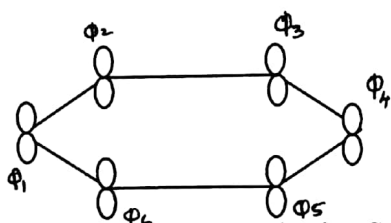
and evaluated for energy. Different energy states thus calculated and corresponding wave functions are given below.

ϵ_4 —————	$\alpha - 1.618\beta$	$\phi_4 = 0.372 \phi_1 - 0.602 \phi_2 + 0.602 \phi_3 - 0.372 \phi_4$
ϵ_3 —————	$\alpha - 0.618\beta$	$\phi_3 = 0.602 \phi_1 - 0.372 \phi_2 - 0.372 \phi_3 + 0.602 \phi_4$
ϵ_2 — $\uparrow\downarrow$ —	$\alpha + 0.618\beta$	$\phi_2 = 0.602 \phi_1 + 0.372 \phi_2 - 0.372 \phi_3 - 0.602 \phi_4$
ϵ_1 — $\uparrow\downarrow$ —	$\alpha + 1.618\beta$	$\phi_1 = 0.372 \phi_1 + 0.602 \phi_2 + 0.602 \phi_3 + 0.372 \phi_4$



Plot of wave function and energy of butadiene.

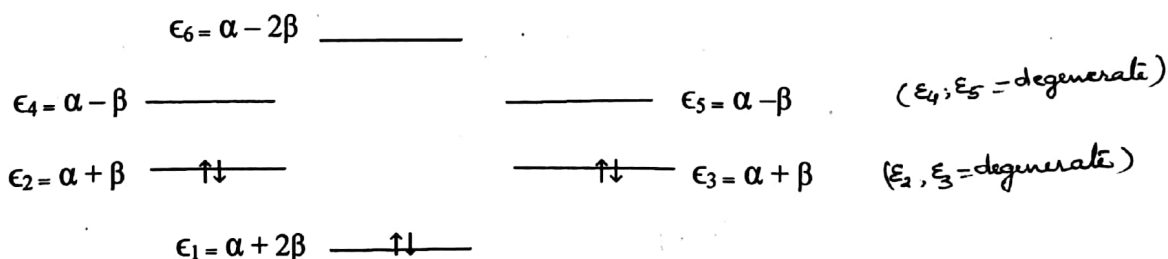
Pi molecular orbitals of Benzene



Trial function

$$\phi = C_1\phi_1 + C_2\phi_2 + C_3\phi_3 + C_4\phi_4 + C_5\phi_5 + C_6\phi_6$$

Energy values calculated using Schrodinger equation are as given below.



Resonance energy of benzene is calculated as 2β

$$\phi_5 = \sqrt{\frac{1}{12}} (\phi_1 + \phi_2 - 2\phi_3 + \phi_4 + \phi_5 - 2\phi_6), \quad \phi_6 = \sqrt{\frac{1}{6}} (\phi_1 - \phi_2 + \phi_3 - \phi_4 + \phi_5 - \phi_6)$$

$$\phi_4 = \frac{1}{2} (\phi_1 - \phi_2 + \phi_4 - \phi_6)$$

$$\phi_3 = \sqrt{\frac{1}{12}} (\phi_1 - \phi_2 - 2\phi_3 - \phi_4 + \phi_5 + 2\phi_6)$$

$$\phi_2 = \frac{1}{2} (\phi_2 + \phi_3 - \phi_5 - \phi_6)$$

$$\phi_1 = \sqrt{\frac{1}{6}} (\phi_1 + \phi_2 + \phi_3 + \phi_4 + \phi_5 + \phi_6)$$

Aromaticity - The Huckel ($4n+2$) rule of Aromaticity.

The molecular orbital theory (MOT) points out special electronic stability of ring compounds with $n\pi$ -electrons where n is an integer 1,2,3....

Huckel rule states that monocyclic, coplanar ring compounds containing $(4n+2)$ π electrons possess high stability and exhibits aromaticity. Thus, to be *aromatic* a molecule must have $2(n=0), 6(n=1), 10(n=2), \dots \pi$ electrons. Such molecules will exhibit aromatic character such as substitution reactions instead of addition reactions. Thus Cyclopropenyl salts ($n=0$), benzene ($n=1$), naphthalene ($n=2$) etc.. are aromatic.
